Urban green areas in The Hague



Het Plein, The Hague. Photo by Zairon, via Wikimedia Commons, CC BY-SA 4.0.

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Spatial Data Visualization and Analysis

NOVA IMS
INFORMATION MANAGEMENT SCHOOL

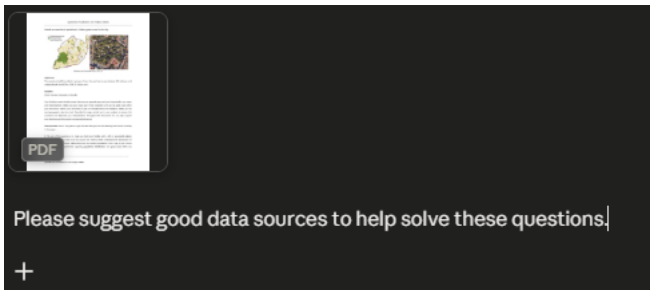
Hands-on exercise (4 questions) – Urban green areas in the city.

Introduction:

For this exercise, I chose The Hague, a big and busy city close to my hometown. In the Netherlands, green spaces are considered to be very important, because they are good for people's health and make cities nicer places to live. In this report, I look at how parks and other green areas are spread across the neighbourhoods of The Hague, and whether there is a link between where people live and how much green space they have nearby. I use open data from CBS (neighbourhoods and population) and the Municipality of The Hague (green areas and dunes).

Question 1:

For the first question, I wanted to create an overview of the distribution of green areas across The Hague. Before collecting the data, I used AI (Claude) to help identify possible spatial data sources.



The suggested sources were then checked against official data portals before being used in the analysis. I initially considered datasets such as the BGT and the municipal tree register. However, after testing different options in ArcGIS Pro, I decided to use the official green space datasets provided by the Municipality of The Hague because they gave a clearer representation of the different types of green areas relevant to this map.

The municipal boundary was obtained from the CBS Wijken en Buurten 2025 dataset through the PDOK Web Feature Service (WFS) [1]. This dataset contains the geometries of Dutch municipalities, districts and neighbourhoods together with statistical attributes. In ArcGIS Pro, I selected the municipality of 's-Gravenhage (GM0518). The dataset separates the land and water parts of the municipality. I therefore used the feature with water = NEE as the municipal land boundary and the feature with water = JA to represent the municipality's marine area in the North Sea.

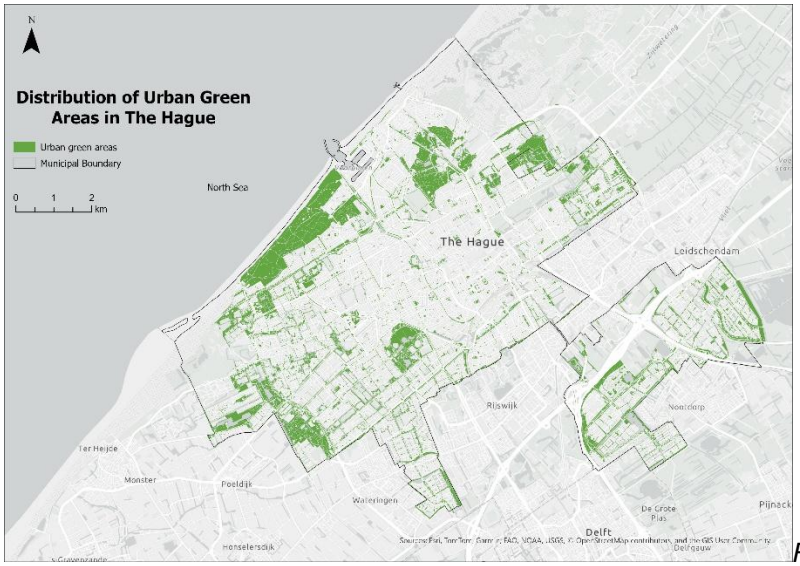
For the detailed green areas, I used the Municipality of The Hague's Groenvlak layer from the *Natuur en landschapsbeheer* ArcGIS service [2]. This polygon dataset contains different types of managed green space and vegetation across the city, including grass, shrubs, hedges, woodland and other forms of vegetation. Because the purpose of this first map was to show the overall distribution of green rather than individual maintenance categories, these features were displayed together as Urban green areas.

I added the trees located on the street but I chose to remove them because they do not provide relevant information for green areas in my opinion because I want the map to show the

distribution of green areas across The Hague. The tree dataset contains a very large number of individual points. When I show all of them on a city-wide map, the map becomes very busy and the tree points take too much attention. This makes it harder to see the larger green areas, dune areas and the overall pattern of green space in the city. The data is taken from Gemeente Den Haag, Straatboom / Bomenbeheer. [8]



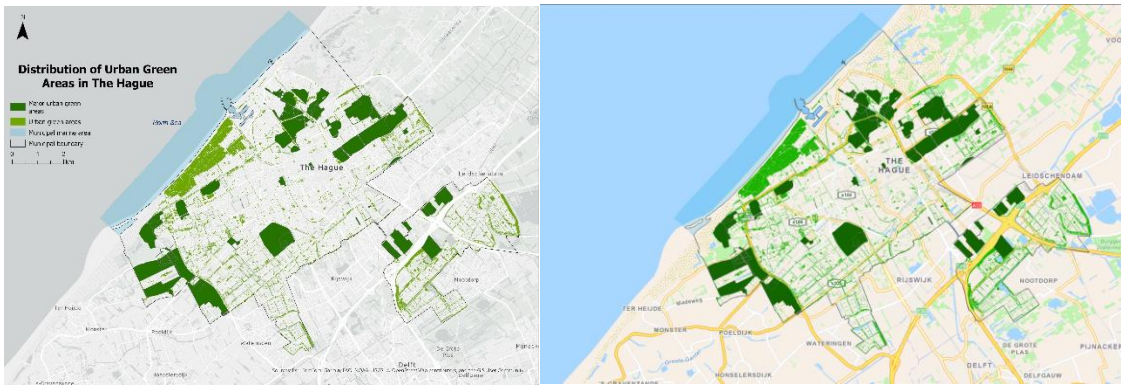
Map with all the trees on the street (too busy).



First draft

I also added the larger green areas from the Municipality of The Hague's Stedelijke Groene Hoofdstructuur (SGH). More specifically, I used the layer *Grote groengebieden en overig groen SGH* [3]. This dataset represents the larger green areas that are part of the city's strategic green structure. I displayed these as Major urban green areas, using a darker shade of green so that they could be distinguished from the finer network of managed urban green space.

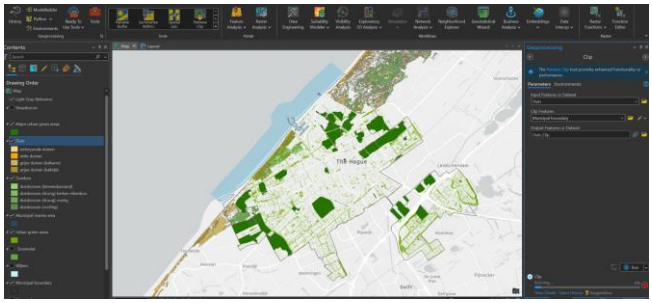
During the mapping process, I noticed that conventional basemaps showed substantial green areas along the coast that were not fully represented by these two layers. Since the coastal landscape is an important part of The Hague, I decided to include the dune environment explicitly.



My map with light grey base map *vs* *Navigation base map which shows more green*

For this, I used the Municipality of The Hague's Duin dataset [4] and Duinbos dataset [5]. The original dune dataset distinguishes several dune habitat types, while the dune woodland dataset contains different forms of woodland associated with the dune environment. For this overview map, I simplified these classifications visually into two categories: Dune areas and Dune woodland.

Only the dune features located within The Hague were required. I therefore used the Clip tool in ArcGIS Pro, with the municipal land boundary as the clipping feature. This removed dune and dune-woodland features outside the municipality while retaining those located within The Hague. The original classifications were preserved in the data, but a single symbol was used for each category to keep the final map readable.



Clip tool

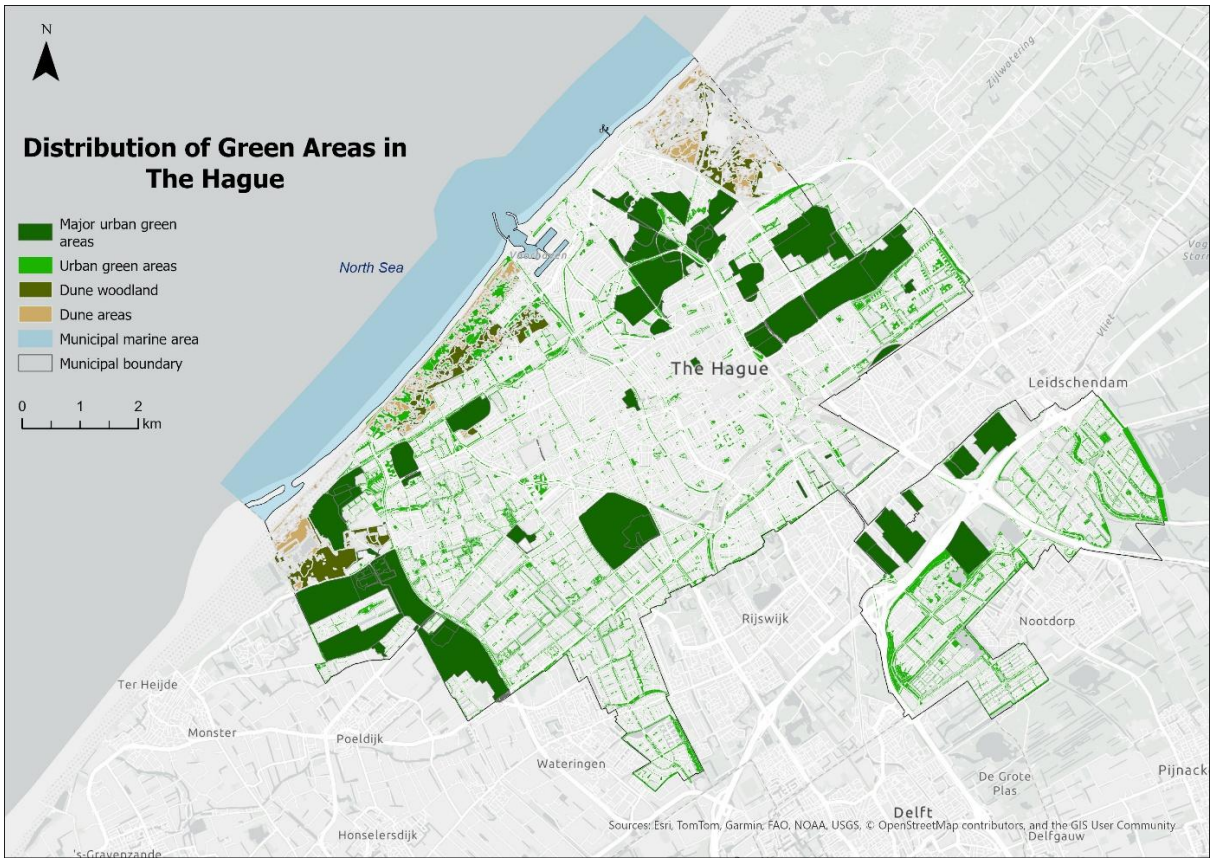
I considered the inclusion of the dunes important because the Dutch coastal landscape has both ecological and protective functions. Beaches and dunes form an important part of the Netherlands' coastal protection system, while also providing areas for nature and recreation [6]. Their inclusion therefore gives a more complete representation of the green landscape of The Hague. This may also become relevant in the later questions when comparing the distribution of green space with the distribution of the population.

For the background, I tested several ArcGIS basemaps. More detailed basemaps contained many roads, street names, place names and additional colours. This made the map visually busy and reduced the emphasis on the green-space layers. I therefore selected the ArcGIS Light Gray basemap [7]. Its neutral appearance allows the thematic green-space data to remain the main visual focus of the map.

Finally, the municipal marine area was displayed in light blue and labelled North Sea. I added this mainly as geographic context. Readers who are not familiar with The Hague can

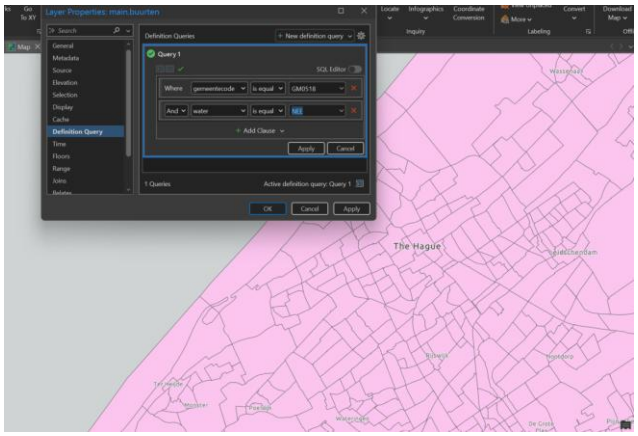
immediately see that the city is located directly on the coast, without requiring a more detailed basemap.

The final map therefore distinguishes six elements: Major urban green areas, Urban green areas, Dune woodland, Dune areas, Municipal marine area and the Municipal boundary. Together, these layers provide a representation of green spaces in The Hague.



Final result Q1

Question 2: Green coverage per neighbourhood


Adding the urban administrative units.

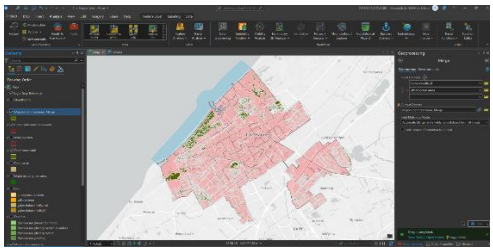
For this map, I used the neighbourhood level (buurt) from the CBS Wijk- en Buurtkaart 2024. [9] This is the smallest administrative unit in the Netherlands for which official population data is

available. I selected only the neighbourhoods of The Hague (municipal code GM0518) and removed neighbourhoods that consist only of water. This left 114 neighbourhoods.

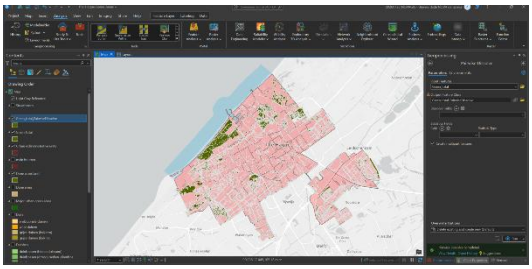
As green areas, I used the Groenvlak layer of the Municipality of The Hague [2], shown as Urban green areas in Question 1, together with the dune woodland layer [5]. I did not include the major green areas (SGH) [3] in the calculation. These polygons outline whole parks, including paths, ponds and playgrounds, which would make the green percentages too high. The Groenvlak layer already contains the green surfaces inside these parks.

I did the data processing in ArcGIS Pro in the following steps:

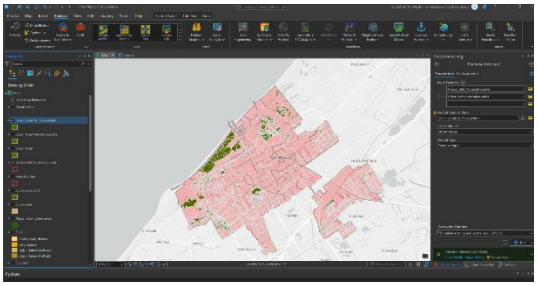
1. **Merge**: I combined the green layers into one layer.



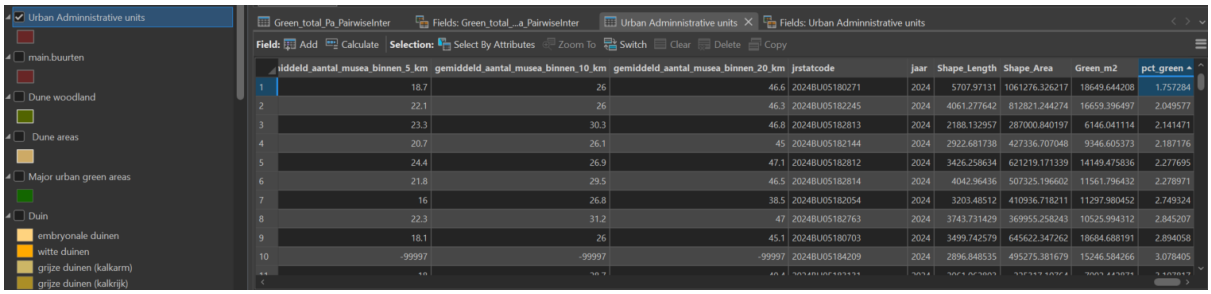
2. **Pairwise Dissolve**: I joined overlapping green areas together, so that no area was counted twice.



3. **Pairwise Intersect**: I cut the green layer along the neighbourhood boundaries, so every piece of green belongs to one neighbourhood.



4. I added the green area (m^2) per neighbourhood to the neighbourhood layer with **Join Field**, using the neighbourhood code (buurtcode).

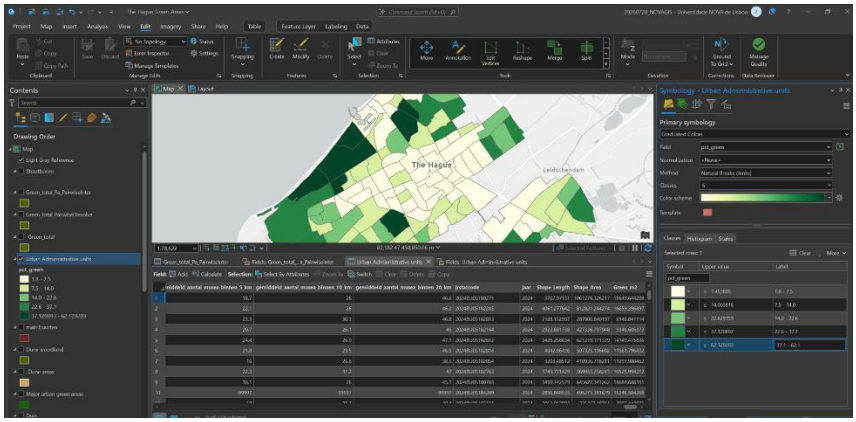


5. Finally, I calculated the green coverage for each neighbourhood:

Green coverage (%) = green area in neighbourhood (m^2) / total area of neighbourhood (m^2) × 100

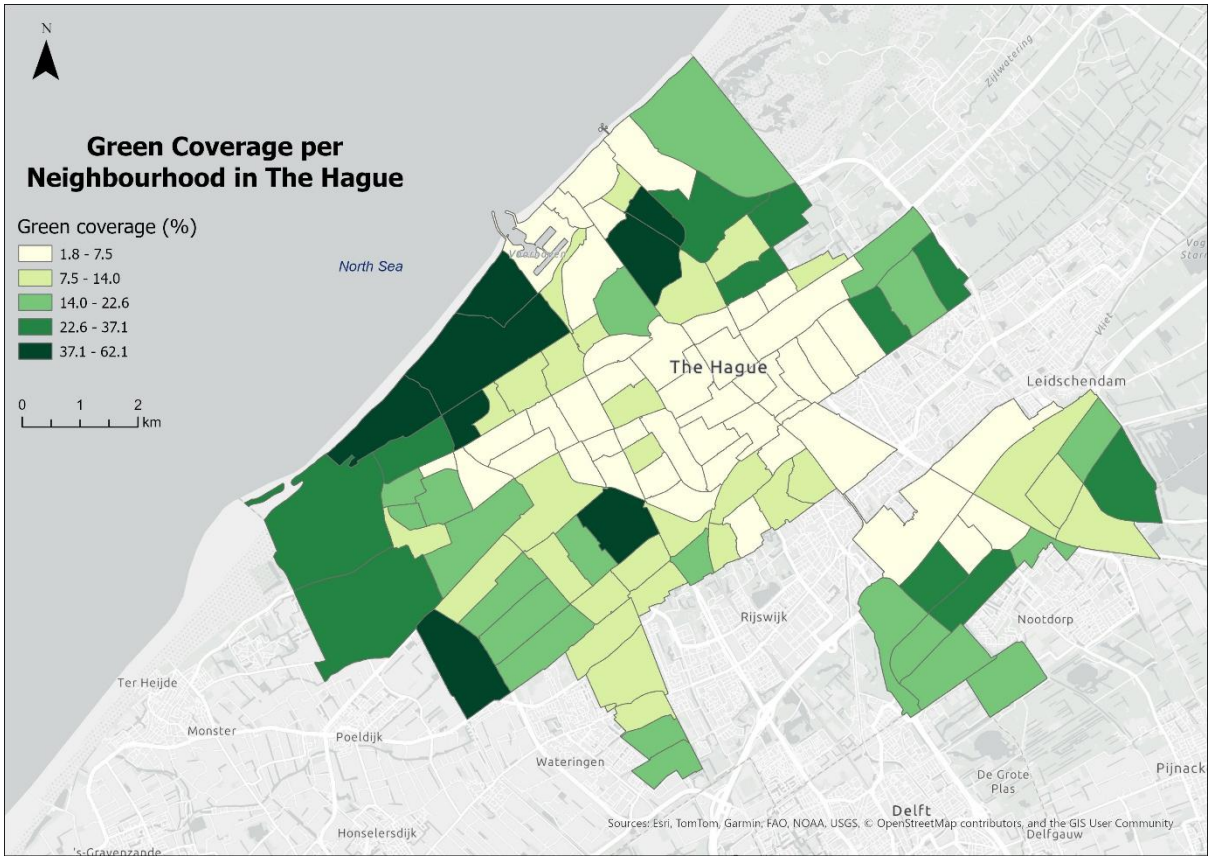
I chose a percentage instead of a count or total area because neighbourhoods have very different sizes. A large neighbourhood would automatically have more green in total, even if it is not greener. A percentage corrects for this, so neighbourhoods can be compared fairly.

For the classification, I used Natural Breaks (Jenks) with five classes. The values range from 1.8% to 62.1%, but most neighbourhoods have low values and only a few are very green (see histogram). Natural Breaks groups neighbourhoods with similar values and places the class



boundaries at natural gaps in the data. With Equal Interval, most neighbourhoods would fall into the lowest one or two classes, and the differences between them would disappear. I used a light-to-dark green colour scheme: light means little green, dark means a lot of green.

The map shows a clear pattern. Most neighbourhoods in the centre of The Hague have less than 15% green coverage. The highest values are found along the coast and at the edges of the city, near the dunes, woods and large parks such as the Scheveningse Bosjes and Clingendael.

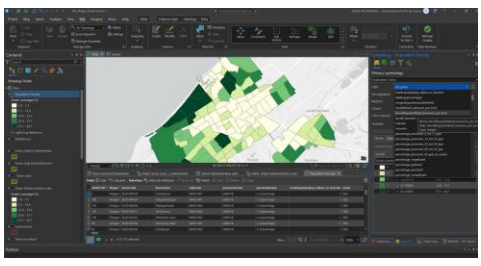


Final result Q2

Limitations: The percentage is calculated using the total area of each neighbourhood. For some coastal neighbourhoods this also includes beach, harbour or water, which makes their percentage slightly lower. In addition, open dune areas and private gardens are not included, so the real amount of green is somewhat higher than the map shows.

Question 3: Population density by neighbourhood

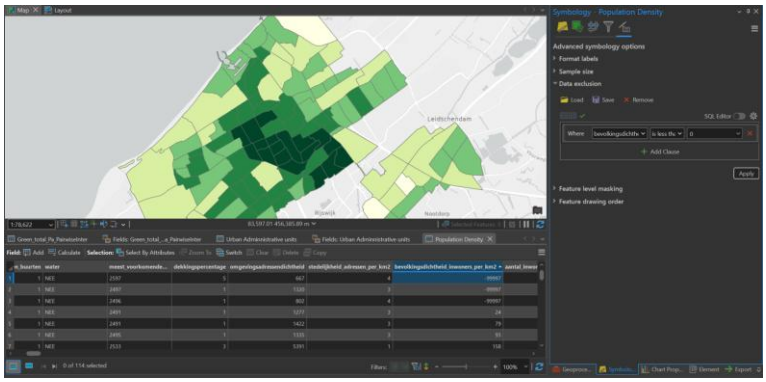
For this map, I used the same 114 neighbourhoods (*buurten*) from the CBS Wijk- en Buurtkaart 2024 [9] as in Question 2, so the two maps can be compared directly. I made a copy of the neighbourhood layer in ArcGIS Pro, so the Question 2 map stayed unchanged.



Made a copy and changed the field to population density per km^2.

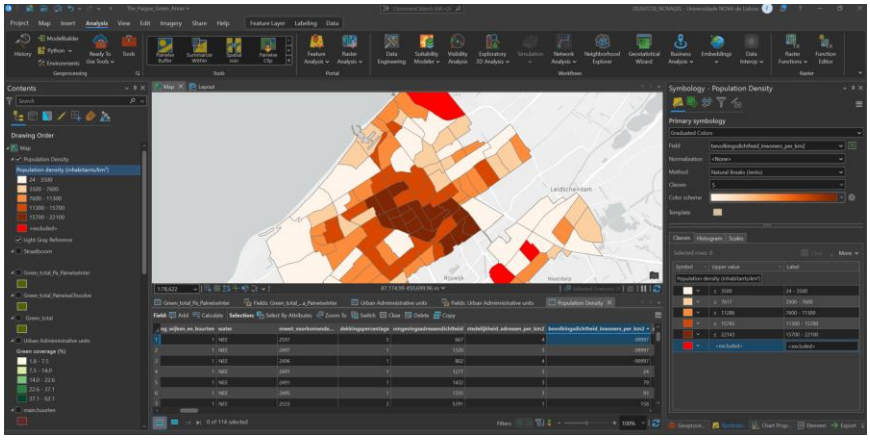
I mapped population density (inhabitants per km^2) instead of the total number of inhabitants. The neighbourhoods differ a lot in size, and a large neighbourhood can have many residents simply because it covers more land. Density corrects for this, so the neighbourhoods can be compared fairly. I used the field *bevolkingsdichtheid_inwoners_per_km2*, which CBS calculates as the number of registered inhabitants divided by the land area of the neighbourhood. Because only land area is used, water and sea do not affect the values.

Three neighbourhoods had the value -99997. CBS uses this value when the number of residents is too small to publish for privacy reasons. If I had kept these values, they would have created a false extra class and distorted the classification. I therefore excluded them using the *Data exclusion* option in the symbology settings. They are not shown on the map.



Three neighbourhoods were excluded because they had the value -99997, which CBS uses for unknown or confidential data. Including them would have strongly distorted the classification.

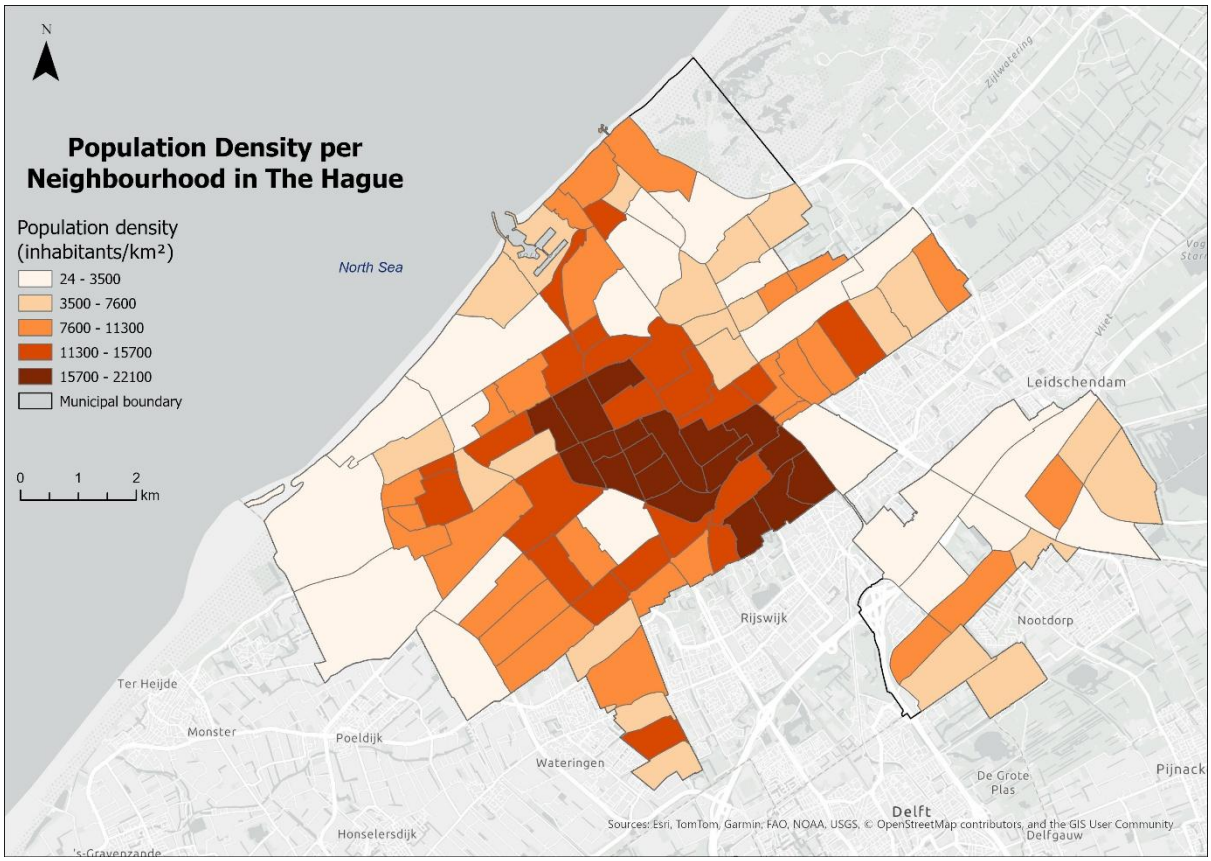
For the classification, I used Natural Breaks (Jenks) with five classes, the same method and number of classes as in Question 2. The values range from 24 to 22,143 inhabitants per km^2. Natural Breaks places the class boundaries at natural gaps in the data, so neighbourhoods within one class are as similar as possible (see histogram). I used an orange colour scheme, so this map is clearly different from the green map of Question 2: light means low density, dark means high density.



Rounded class values and final colour scheme.

The map shows a clear pattern. The highest densities are found in the centre of The Hague and the older neighbourhoods just south of it, where many small homes are built close together. The

lowest densities are found at the edges of the city, in neighbourhoods with large parks, dunes, woods or business areas, where few people live on a large area.

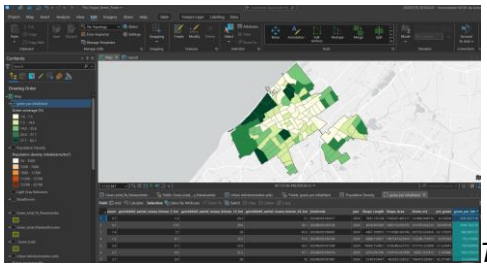


Final Result Q3

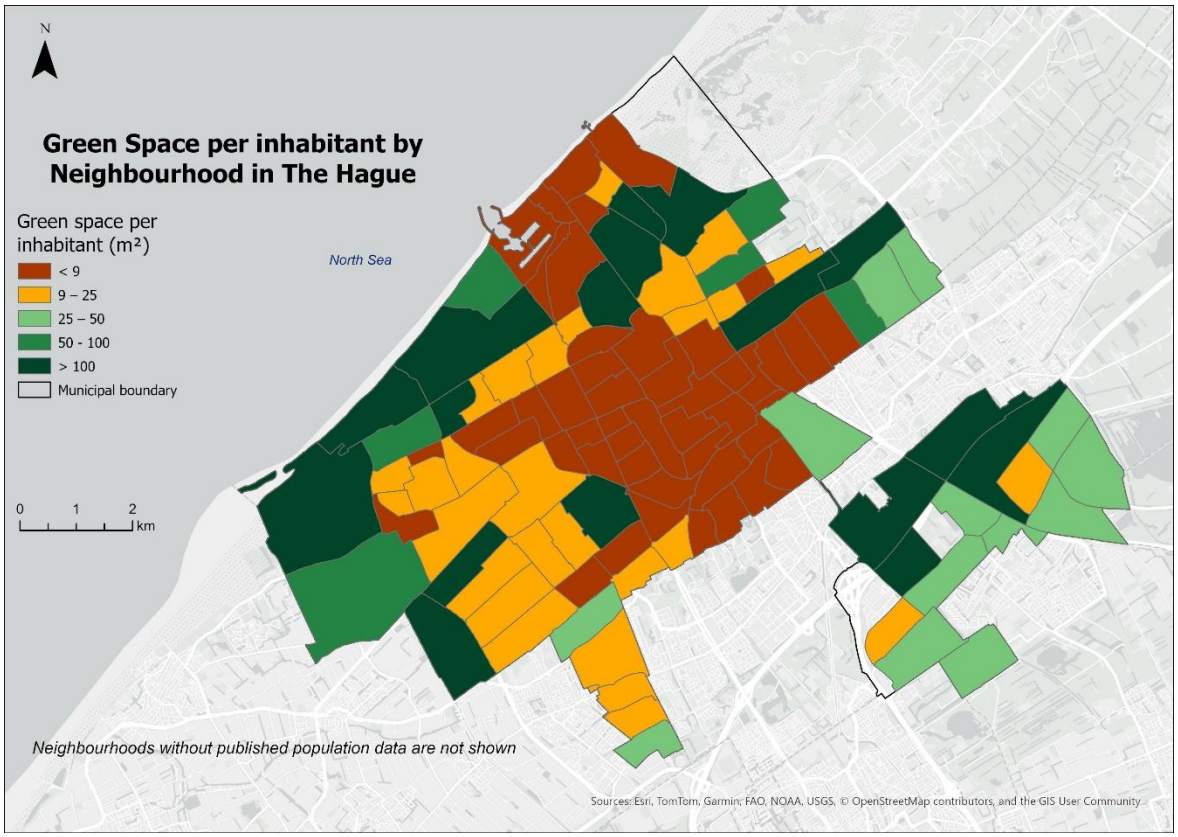
Limitations: Three neighbourhoods have no published data, so they are missing from the map. Also, the density is an average for the whole neighbourhood; within a neighbourhood, some streets can be much denser than others.

Question 4 - Green space per inhabitant

To show the relationship between green space and residents, I calculated the amount of green space per inhabitant for each neighbourhood. I made a new copy of the neighbourhood layer and added a field *green_per_inh*. First, I selected only neighbourhoods with a population above zero, which left out the three neighbourhoods without published data (-99997). For these 111 neighbourhoods, I divided the green area from Question 2 (*Green_m2*) by the number of inhabitants (*aantal_inwoners*). The result is shown in square metres of green per inhabitant (m^2).



The calculated green space per inhabitant.



For this map, I used manual class boundaries instead of Natural Breaks. A few neighbourhoods with large parks and very few residents, such as Zuiderpark, have more than 3,000 m^2 of green per inhabitant. With Natural Breaks, almost all other neighbourhoods would fall into the lowest class. The idea to use 9 m^2 as the lowest boundary came from AI (Claude), which described it as a guideline often attributed to the World Health Organization. When I checked this, I found that the value is widely used in research as a minimum amount of green space per person [10], but

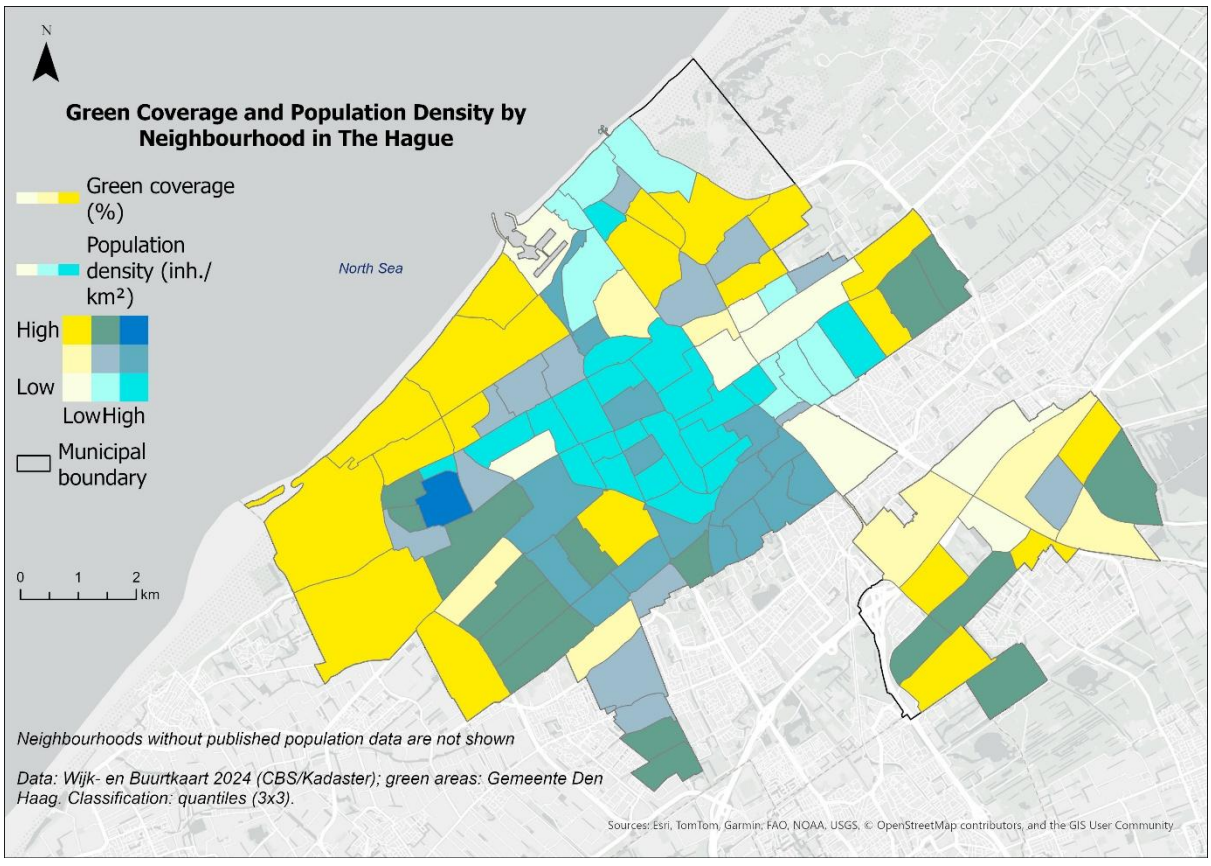
that no official WHO publication with this number could be found. I therefore use 9 m^2 only as a reference value, not as an official standard.

The map shows that most neighbourhoods in the centre of The Hague have less than 9 m^2 of green space per inhabitant. These are surrounded by neighbourhoods with 9 to 25 m^2. Neighbourhoods near the dunes, woods and large parks at the edges of the city have much more, often over 100 m^2 per inhabitant.

$$\text{Green space per inhabitant} = \frac{\text{green area in the neighbourhood } (m^2)}{\text{number of residents}}$$

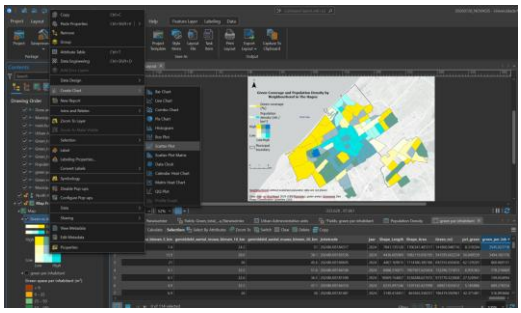
To show both variables in one map, I made another copy of the neighbourhood layer and used the Bivariate Colors symbology in ArcGIS Pro. I combined green coverage (*pct_green*, from Question 2) with population density (*bevolkingsdichtheid_inwoners_per_km2*, from Question 3). Both variables were divided into three classes (low, medium, high) using quantiles, which gives nine colour combinations. I chose quantiles so that each class contains the same number of neighbourhoods and all nine colours are used. The three neighbourhoods without population data (-99997) were removed with a definition query, so they do not affect the classes.

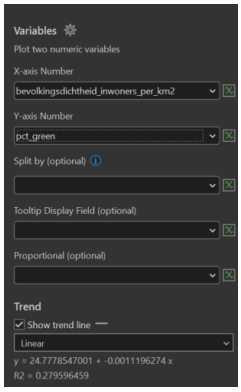
The map shows a clear pattern. Turquoise neighbourhoods, with high population density and little green, are concentrated in the centre of The Hague. Yellow neighbourhoods, with low density and a lot of green, are found along the coast and at the edges of the city. Only a few neighbourhoods, shown in dark blue, combine high density with a lot of green.



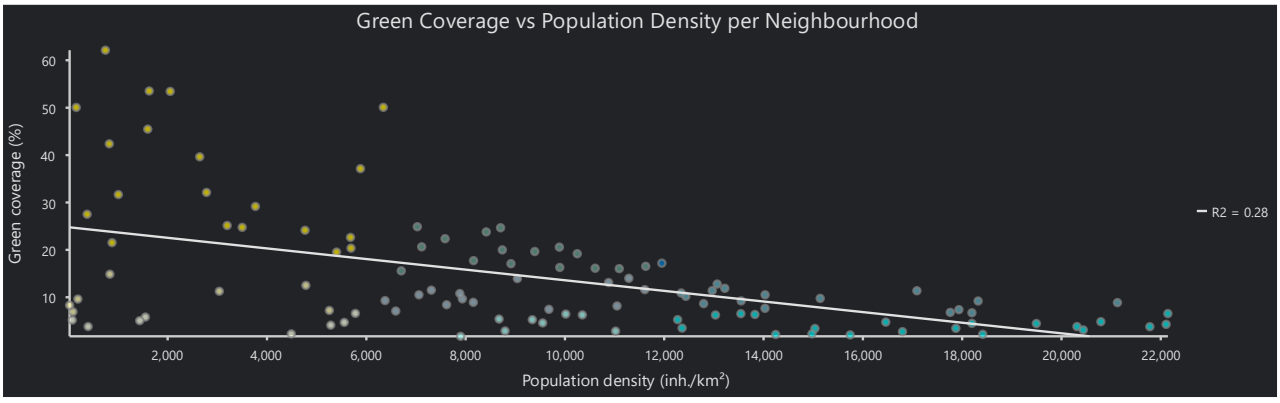
Bivariate map combining green coverage and population density per neighbourhood (quantiles, 3 × 3): turquoise shows high density with little green, yellow shows low density with a lot of green.

I created a scatter plot because the maps show the spatial pattern, but not how strong the relationship between population density and green coverage is.





Values used to create scatter plot



Scatter plot of population density and green coverage for 111 neighbourhoods in The Hague, with linear trend line $R^2 = 0.28$.

To measure the strength of the relationship, I made a scatter plot of population density and green coverage for the 111 neighbourhoods with population data. The trend line goes down, which shows a negative relationship: neighbourhoods with a higher population density tend to have less green space. The R^2 is 0.28, which means that about 28% of the differences in green coverage between neighbourhoods is related to population density. This is a weak to moderate

relationship. The other 72% may be related to other factors, such as the age of the neighbourhood, its location near the coast or dunes, and the presence of large parks. The relationship is clear enough to confirm the pattern on the maps, but population density alone does not explain how much green a neighbourhood has.

Limitations: As in the other maps, only green space within the neighbourhood boundary is counted, while residents can also use parks in nearby neighbourhoods. Street trees and private gardens are not included. The results also depend on the chosen neighbourhood boundaries, and three neighbourhoods without published population data are missing.

Conclusion

All three results point in the same direction. Map 1 shows that residents in the centre have the least green space per person, the bivariate map shows that these are also the most densely populated neighbourhoods, and the scatter plot confirms a negative relationship between population density and green coverage ($R^2 = 0.28$). This is a common pattern in large cities: historic centres were built densely, with little space left for parks, while newer neighbourhoods and areas at the edges have more room for green. The relationship is weak to moderate, so density is not the only factor, but the pattern is clear. For the city council, the dense central neighbourhoods with less than 9 m^2 of green per inhabitant should have priority when planning new green spaces.

Use of AI

During this assignment, I used two AI tools; Claude and ChatGPT. It helped me find possible data sources, guided me through some steps in ArcGIS Pro and suggested the 9 m^2 reference value. All GIS work was done by me, including downloading, clipping, merging and analysing the data, and designing the maps. The final choices about the data, classification methods and interpretation of the results were my own.

References

[1] CBS / PDOK — Wijken en Buurten 2025
Municipal boundary and municipal marine area.
CBS Wijken en Buurten via PDOK

[2] Gemeente Den Haag — Groenvlak, Layer 14
Detailed managed urban green-space polygons.
Groenvlak – Gemeente Den Haag

[3] Gemeente Den Haag — Grote groengebieden en overig groen SGH, Layer 79
Major green areas within the *Stedelijke Groene Hoofdstructuur*.
Major green areas SGH – Gemeente Den Haag

[4] Gemeente Den Haag — Duin, Layer 39
Dune habitat areas.
Duin – Gemeente Den Haag

[5] Gemeente Den Haag — Duinbos, Layer 37
Dune woodland habitat areas.
Duinbos – Gemeente Den Haag

[6] Rijkswaterstaat — Coastal protection and dune management
Background information on the protective role of beaches and dunes along the Dutch coast.
Rijkswaterstaat – Coastal protection and maintenance

[7] Esri — ArcGIS Light Gray basemap
Neutral reference basemap used as background for the thematic map.
ArcGIS Light Gray basemap

[8] **Gemeente Den Haag — Straatboom / Bomenbeheer**
Official municipal tree-management GIS layer.
Street trees – Municipality of The Hague

[9] Centraal Bureau voor de Statistiek (CBS) & Kadaster. (2025). *Wijk- en Buurtkaart 2024, versie 2* [Dataset]. https://www.cbs.nl/nl-nl/dossier/nederland-regionaal/geografische-data/wijk-en-buurtkaart-2024.

[10] Urban Green Space per Capita for Sustainable and Equitable Urban Planning: A Systematic Review and Bibliometric Analysis. (2025). *Land*, 15(1), 29. https://www.mdpi.com/2073-445X/15/1/29.